Paper 09 - Hamiltonian Window Emergence

CQE-paper-09

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

Describe local inside global temporal emergence as Hamiltonian-window readout over transported states.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- promoted formal paper: production\formal-papers\CQE-paper-09\FORMAL_PAPER.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-09\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-09\verify_hamiltonian_window_emergence.py
- receipt: production\formal-papers\CQE-paper-09\hamiltonian_window_emergence_receipt.json (Hamiltonian window emergence: pass)
- body: production\papers\CQE-paper-09\PAPER-BODY.md
- source: production\papers\CQE-paper-09\SOURCE.md
- formal: production\papers\CQE-paper-09\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-09\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-09\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-09.md

Paper 09 - Hamiltonian Window Emergence

Abstract

Paper 09 formalizes temporal emergence as an iterated local-window read over already transported centers. The theorem is finite and exact: given an ordered list of carried centers, a Hamiltonian window of width `w` reads a contiguous slice forward, reads the same slice backward as a well-formed reverse receipt, and emits a composite center that preserves the source-center sequence. The first production instance uses widths `3`, `5`, and `7`, corresponding to the 1-3, 1-5, and 1-7 bar reads. Starting with six base centers, the iteration emits four order-2 windows, three order-3 windows, and two order-4 windows.

This paper proves the local temporal-construction mechanism. It does not claim that the static four-frame Z4 label template by itself determines the Rule 30 temporal trace; the temporal Z4 verifier records that as a boundary.

Proof/Exposure Hierarchy

The proof-carrying content of this paper is the mathematics: the definitions, lemmas, constructions, examples, and receipts that establish the claimed result. Paper 00, hand routes, analog tools, workbook language, and obligation ledgers are supplemental validation and exposure layers. They exist to make the math inspectable, reproducible, and accessible without requiring a particular software stack. The hand route is not the purpose of the paper; it is a way to expose the same state transitions with ordinary marks, tokens, lines, or any equivalent physical substitute.

Definitions

A **center state** is a pair `(paper_id, center)` where `center` is the surviving local quantity carried from that paper.

A **Hamiltonian window** is a contiguous slice of fixed odd width `w` over an ordered center-state sequence.

A **forward read** preserves the source order:

```
C_i -> C_{i+1} -> ... -> C_{i+w-1}
```

A **backward read** records the reverse receipt:

```
C_{i+w-1} <- ... <- C_{i+1} <- C_i
```

A **surviving composite center** is the ordered join of every center in the window. It is accepted only when the forward and backward receipts contain the same source centers in opposite order.

Main Claim

Theorem 9.1, Hamiltonian Window Emergence. For any finite ordered sequence of center states and any fixed window width `w <= n`, the Hamiltonian window read emits exactly `n - w + 1` surviving composite centers. Each composite center preserves its source centers, source indices, forward receipt, and backward receipt. Iterating widths `3`, `5`, and `7` over the CQECMPLX base centers produces the order counts `4`, `3`, and `2`.

Proof

Let `S = [s_0, ..., s_{n-1}]` be a finite ordered sequence and let `w <= n`. The valid window starts are:

```
0, 1, ..., n - w
```

There are therefore `n - w + 1` windows. For each start index `i`, define the window slice:

```
W_i = [s_i, s_{i+1}, ..., s_{i+w-1}]
```

The forward read records the centers of `W_i` in source order. The backward read records the same centers in reverse order. Since reversal is an involution, reversing the backward read recovers the forward center sequence. Thus the emitted composite center preserves the source centers and their provenance.

For the CQECMPLX production instance, begin with six base center states. Width `3` produces `6 - 3 + 1 = 4` order-2 reads. Appending the resulting order-2 state gives seven states, so width `5` produces `7 - 5 + 1 = 3` order-3 reads. Appending that state gives eight states, so width `7` produces `8 - 7 + 1 = 2` order-4 reads. QED.

Relation to Earlier Papers

Papers 01-08 establish carrier, correction, coordinate, repair, path, causal, bridge, and closure-template layers. Paper 09 adds temporal construction: the ordered global object is read from local center windows rather than assumed as an external timeline.

```
local centers
-> width-3 reads
-> width-5 reads
-> width-7 reads
-> composite temporal states with receipts
```

Falsifier

The verifier rejects these overclaims:

```
"the static Z4 chart template proves the temporal trace is period 4"
"a reverse receipt proves physical time reversal"
"a composite center is valid without source-window provenance"
```

Paper 09 proves window emergence and receipt-preserving reverse readability. It does not claim that static chart symmetry alone proves temporal dynamics.

Code Reconstruction

The production verifier is:

```
production/formal-papers/CQE-paper-09/verify_hamiltonian_window_emergence.py
```

It verifies:

```
1. Width-3 reads over six base centers produce four surviving centers.
2. Width-5 reads after appending order 2 produce three surviving centers.
3. Width-7 reads after appending order 3 produce two surviving centers.
4. Every surviving center carries source indices and source centers.
5. Every backward receipt reverses to the forward receipt.
6. The temporal Z4 scope verifier records static-template-only status.
```

Open Audit Boundaries

- The reverse read is a receipt-level reverse, not a proof of physical time reversal.
- The static Z4 chart template does not force the Rule 30 temporal trace.
- Higher-order convergence beyond width 7 is not claimed by this paper.
- Any Hamiltonian or physical dynamics assigned to the composite centers require a later theorem.

Conclusion

Paper 09 proves that temporal structure can be built from local carried centers by finite Hamiltonian-window reads. The proof is the window arithmetic and provenance preservation. The supplemental hand/workbook route merely exposes the same construction; the mathematical result is the receipt-bearing emergence of composite centers from local windows.